



مدارس حديد عز
للتكنولوجيا التطبيقية
ezzsteel applied technology schools

UNIT 1 - Occupational Health, Safety and Environment

LESSON 1 - Foundations of Occupational Health and Safety

Subtopic 1.1 – What is Occupational Health and Safety?

Eng. Nada Ahmed Ali Attia

04/10/2026

Academic Year: 2026/2027
Semester : One

WARM-UP

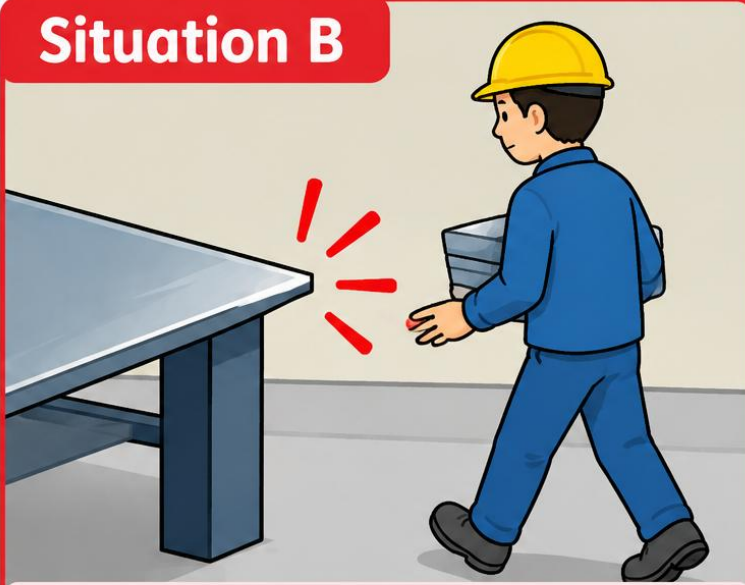
SAME HAZARD, DIFFERENT RISK

Situation A



✓ A sharp edge on a metal sheet is positioned behind a suitable barrier. The area is controlled, and workers cannot normally reach the edge.

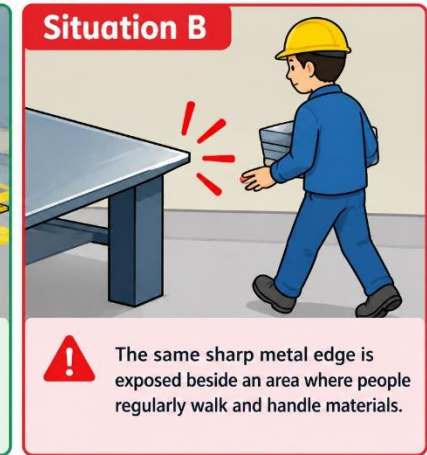
Situation B



⚠ The same sharp metal edge is exposed beside an area where people regularly walk and handle materials.

WARM-UP

SAME HAZARD, DIFFERENT RISK



- Discuss the following questions with a partner:
 1. Has the hazard itself changed?
 2. What is different between the two situations?
 3. In which situation is contact with the sharp edge more likely?
 4. In which situation could the risk be greater?
 5. What action could prevent someone from being exposed?

By the end of this lesson, I will be able to...

1. Define hazard, risk and prevention

Distinguish a hazard/unsafe condition from the possible harm and from a preventive measure.

Distinguish a hazard from a risk

Explain why prevention is required before work begins

Watch and Discuss



SAFETY



Definition - SAFETY

Safety is a condition in which people are not exposed to unacceptable levels of danger.



OCCUPATIONAL HEALTH

Occupational health
includes your:

physical
well-being

mental
well-being

Social
well-being

OCCUPATIONAL HEALTH

Good occupational-health practice aims to:



OCCUPATIONAL HEALTH



Definition - OCCUPATIONAL HEALTH

Occupational health is concerned with protecting and supporting your physical, mental and social well-being in relation to the work you perform and the conditions in which you perform it.



OCCUPATIONAL HEALTH

THINK ABOUT

A worker finishes a shift without suffering an accident. Does this automatically mean that the workplace was healthy and safe?



OCCUPATIONAL HEALTH

Workplace Well-being



1) Physical well-being

Safe posture, healthy body, and personal protection.



2) Mental well-being

Concentration, calmness, focus, and mental well-being.



3) Social well-being

Teamwork, communication, and supportive interaction with colleagues.



Figure 1.1. Occupational health includes physical, mental and social well-being at work.

Foundations of Occupational Health and Safety

SUMMARY !

Define Safety

Occupational Health: Physical, Mental, Social

Occupational Health Aim

Define Occupational Health

Foundations of Occupational Health and Safety

HOMEWORK!

Student Book : Page no. 174 : section A



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SUBTOPIC 1.2 FROM HAZARD TO RISK

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4/10/2026

Academic Year 2026/2027
Semester One

Learning Objectives: From Hazard to Risk

1.

Define Hazard and Risk



2.

Differentiate Incident vs Accident



3.

Understand Exposure and Harm



4.

Analyze Risk: Likelihood & Severity





Incident

An **incident** is an unplanned event that **could result** in injury, ill health, damage or other loss.



Example:

A worker slips on a wet floor but does not fall or get injured → **Incident**

Accident

An **accident** is an unplanned event that results in injury or ill health.



Example:

A worker slips, falls, and breaks their arm → **Accident**

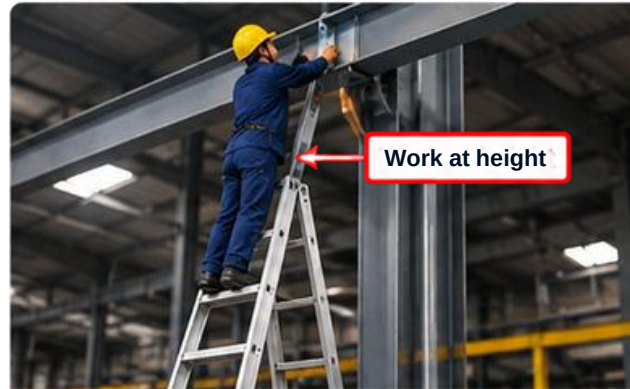
Hazard

A **hazard** is an inherent property or characteristic of an object, substance, activity or situation that has the potential to cause harm to people, property or the environment. A hazard can exist even when nobody is currently being exposed to it.



Sharp metal edge

A sharp metal edge is a hazard that can cause cuts or lacerations.



Ladder or work at height

Working at height is a hazard that can cause a person to fall.



Sulphuric acid

Sulphuric acid is a hazard that can cause chemical burns or respiratory harm.

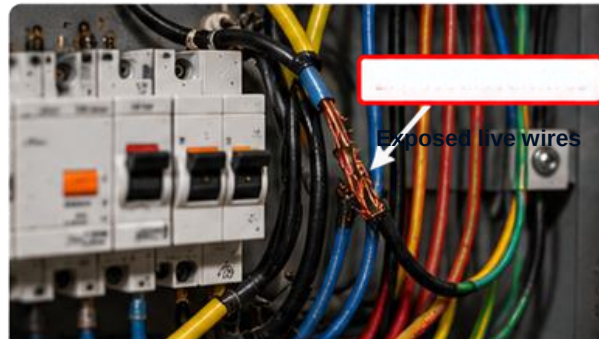
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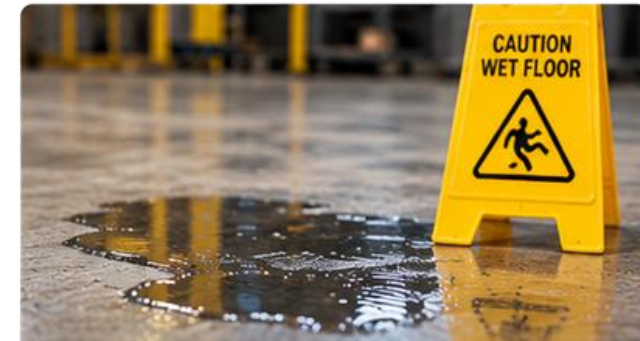
Loose material in a walking

Loose material in a walking route is a hazard that can cause trips, impacts or crushing injury.



Exposed electrical wires

Exposed electrical wires are a hazard that can cause electric shock or burns.



Wet floor

A wet floor is a hazard that can cause slips and falls.

Activity: Identify the Hazard

Look at each picture and write the hazard (only the hazard), not the exposure or harm).



What's the hazard?



What's the hazard?



What's the hazard?



What's the hazard?



What's the hazard?



What's the hazard?



What's the hazard?



What's the hazard?



Answers: Identify the Hazard



What is the hazard?

Hazard: Wet floor



What is the hazard?

Hazard: Exposed electrical wires



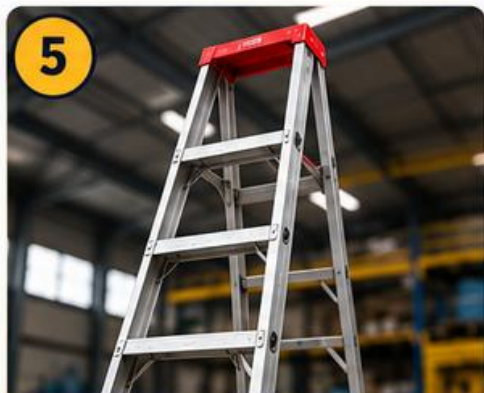
What is the hazard?

Hazard: Sharp metal edge



What is the hazard?

Hazard: Sulphuric acid (chemical substance)



What is the hazard?

Hazard: Ladder / work at height



What is the hazard?

Hazard: Heavy load (manual handling)



What is the hazard?

Hazard: Hot surface (hot metal)



What is the hazard?

Hazard: Loose material in a walking route



Exposure

Exposure describes the condition or event through which a person comes into contact with, or is affected by, a hazard.



Harm

Harm is the injury, ill health, damage or other loss that occurs as a result of exposure to a hazard.

Example 1: Sharp metal edge		Example 2: Work at height		Example 3: Sulphuric acid		Example 4: Loose material in a walking route	
Hazard: Sharp metal edge		Hazard: Ladder or work at height		Hazard: Sulphuric acid		Hazard: Loose material in a walking route	
Exposure: Hand contacts the edge		Exposure: A person loses balance or falls		Exposure: Contact with the substance or inhalation of vapor		Exposure: A person steps on or strikes the material	
Harm: Cut		Harm: Fracture or other fall injury		Harm: Chemical burns or respiratory harm		Harm: Trip, impact or crushing injury	

Hazard → Exposure → Harm



- Identify the Hazard, Exposure and Harm



Look at each picture and decide:

What's the hazard?

What is the exposure?

What is the harm?



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Hazard

Exposure

Harm



Answers: Identify the Hazard, Exposure and

Harm



1

Hazard : Rotating machine part (sharp moving parts)

Exposure e: Hand comes into contact with the rotating part

Harm : Cut or serious injury (e.g. amputation)



2

Hazard : Manual handling (heavy load)

Exposure e: Lifting the heavy box with poor posture

Harm : Back injury (muscle strain)



3

Hazard : Welding (bright light, heat and hot metal)

Exposure e: Exposure to welding arc and hot sparks

Harm : Eye damage (arc eye), burns



4

Hazard : Work at height (fall hazard)

Exposure e: Loss of balance while working on the ladder

Harm : Fracture or other fall injury



5

Hazard : Chemical substance (e.g. acid)

Exposure e: Splash reaches the skin or contact with the liquid

Harm : Chemical burns or skin damage



6

Hazard : Electrical wires (electricity)

Exposure e: Contact with exposed live wires

Harm : Electric shock or burns



7

Hazard : Loose cable in a walking route

Exposure e: Foot catches the cable while walking

Harm : Trip, fall, impact or crushing injury



8

Hazard : Hot surface / hot metal

Exposure e: Hand touches the hot surface

Harm: Thermal burn

What is Risk?

A hazard tells you what has the potential to cause harm.

Risk asks: **how likely** is that harm to occur, and **how serious** could it be?

1. Likelihood

How likely is it that exposure or a harmful event will happen?



More likely = higher risk



2. Severity

How serious could the resulting injury or ill health be?



More severe = higher risk



Risk

The likelihood that a hazard will actually result in harm under particular conditions of use or exposure, taking account of the seriousness of the possible consequence.

Risk depends on **how likely** exposure is and **how serious** the resulting harm could be.

Risk increases when:



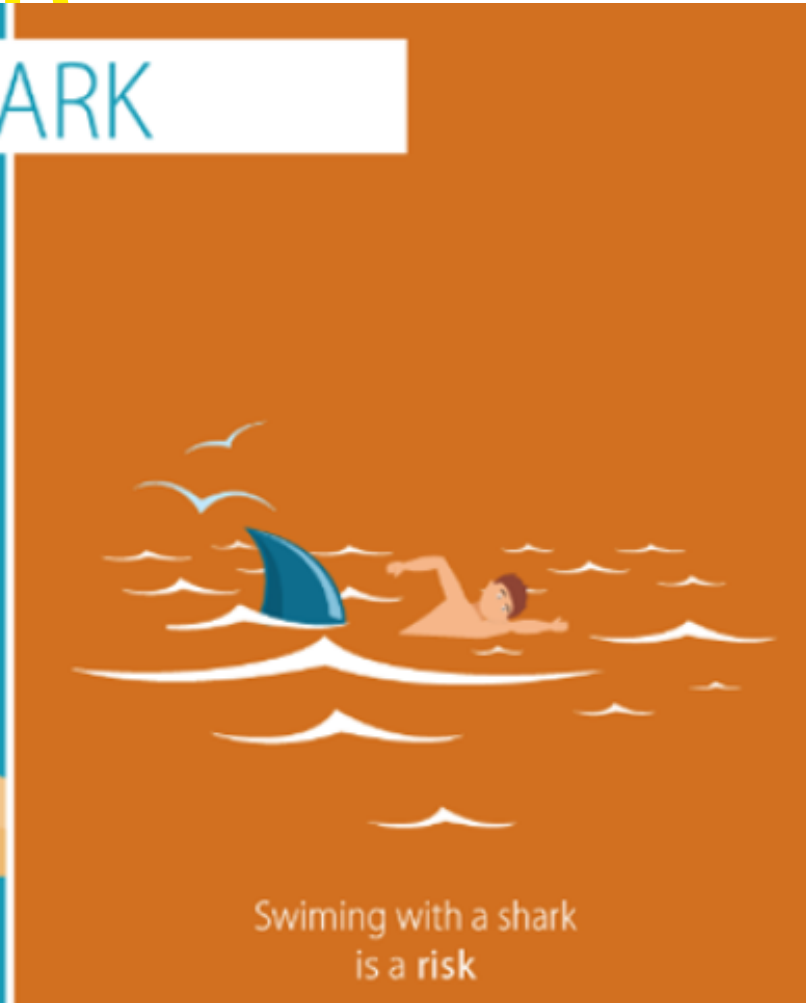
The likelihood of exposure increases.



The severity of the possible harm increases.

Risk VS HAZARD

SHARK



Same Hazard – Different Risk (Practical Example)

The hazard is the same in both cases, but the risk changes because the likelihood of exposure and/or the severity of harm changes.

Condition A – Lower Risk

Chemical solvent stored correctly



- ✓ Closed container
- ✓ Stored correctly
- ✓ Ventilated area
- ✓ Limited access



Hazard
(same in both)

Industrial solvent
(hazardous chemical with harmful vapors)



Exposure

Inhalation of vapors (if container is opened or leaked)

Likelihood
(how likely?)



Low

Unlikely, because exposure is limited.

Severity
(how serious?)



High

Could cause serious respiratory injury (even in small exposure)

Overall Risk:

LOW

Because the likelihood of exposure is low, even though the potential severity is high.

Condition B – Higher Risk

Same chemical solvent, used in poor conditions



- ✗ Open container
- ✗ Being handled
- ✗ Poor ventilation
- ✗ No suitable controls



Hazard
(same in both)

Inhalation of solvent
(hazardous chemical with harmful vapors)



Exposure

Inhalation of vapors (during handling)

Likelihood
(how likely?)



High

Likely, because exposure is frequent and uncontrolled.



High

Could cause serious respiratory injury.

Overall Risk:

HIGH

Because the likelihood of exposure is high and the potential severity is also high.



The hazard can stay the same while the risk changes.

RISK — COMPLETE SUMMARY

Incident

An unplanned event that could have caused harm, but no actual harm



Example: A small solvent spill that was cleaned immediately (no one was injured).

Accident

An unplanned event that actually resulted in harm, damage or loss.



Example: A worker slipped on the spill and broke a hand (actual injury occurred).

Risk

The possibility that harm will occur under the conditions present.

Risk depends on:

Likelihood
(How likely is it to happen?)



Severity
(How serious could it be?)

Hazard

Something that has the potential to cause harm.



Example: Industrial solvent (hazardous chemical with harmful vapors)

Exposure

How a person comes into contact with the hazard, or is affected by it.



Example: Inhalation of vapors (when container is opened or during use)

Harm

The injury, ill health, damage or other loss that occurs as a result of exposure to a hazard.



Example: Respiratory irritation, long-term health problems, or a cut injury.



References

All definitions, examples and key points are taken from the Student Book.



Main Source

Engineering Level 3
Health and Safety in the Engineering Workplace
Student Book



Student Assignment – Risk in Real Life

Apply what you learned about hazard, exposure, harm, likelihood, severity and risk.



Your Task

Choose **at least 4** of the situations below (or use your own real-life examples), then complete a **risk assessment** for each one.



What to do

- 1 Identify the hazard.
- 2 Describe how exposure can happen.
- 3 Describe the possible harm.
- 4 Decide the **likelihood** (Low / Medium / High). Explain your reason.
- 5 Decide the **severity** (Low / Medium / High). Explain your reason.
- 6 Conclude the overall risk (Low / Medium / High).



Success Criteria

- ✓ Clear and correct identification of hazard, exposure and harm.
- ✓ Logical explanation for likelihood and severity.
- ✓ Overall risk is reasonable.
- ✓ Neat and organized presentation (table + short explanation).
- ✓ **Bonus:** Suggest at least one control measure to reduce the risk.

Example situations

1 Industrial solvent



Using a chemical solvent in an open container in a poorly ventilated area.

2 Sharp edge



A sharp edge on a metal sheet.

3 Electrical socket



An overloaded electrical socket.

4 Working at height



A person using a ladder to work at height.

5 Hot surface



A hot cooking surface in a kitchen.

6 Moving machinery



A rotating machine part without a guard.

7 Slippery floor



A wet and slippery floor in a walkway.

8 Sun exposure



Working outdoors under strong sunlight for a long time.



Your Risk Assessment Table

No.	Situation	Hazard (what can cause harm?)	How can exposure happen?	Possible harm (what could happen?)	Likelihood (Low / Med / High) + reason	Severity (Low / Med / High) + reason	Overall risk (Low / Med / High)	Control measure (optional)
1								
2								
3								
4								



Questions to answer (short answers)

- 1 Which situation has the highest risk? Why?
- 2 Which situation has the lowest risk? Why?
- 3 Can the hazard stay the same while the risk changes? Give an example from your table.
- 4 What control measure would be most effective for the highest risk situation?
- 5 What did you learn from this activity?



Submission

You can submit your work as:

- A completed table (Word / PDF / handwritten)
- OR a short presentation (3–5 slides)
- Include pictures and simple explanations.

Student Assignment – Risk in Real Life (With Solutions)

Apply what you learned about hazard, exposure, harm, likelihood, severity, risk, incident and accident.



Your Task

Choose **at least 4** of the situations below (or use your own real-life examples), then complete a **risk assessment** for each one.



What to do

- 1 Identify the hazard.
- 2 Describe how exposure can happen.
- 3 Describe the possible harm.
- 4 Decide the likelihood (Low / Medium / High). Explain your reason.
- 5 Decide the severity (Low / Medium / High). Explain your reason.
- 6 Calculate the overall risk (Low / Medium / High).
- 7 Identify if it is an **incident** or an **accident** (if applicable).
- 8 Suggest at least one control **measure** to reduce the risk.



Key definitions

- ❖ **Hazard:** Something that has the potential to cause harm.
- ❖ **Exposure:** How a person comes into contact with the hazard.
- ❖ **Harm:** The injury, ill health, damage or other loss that could occur.
- ❖ **Risk:** How likely the harm is and how serious it could be.
- ❖ **Incident:** An unplanned event that could have caused harm, but no harm actually occurred.
- ❖ **Accident:** An unplanned event that resulted in harm, damage or loss.

Example situations

1 Industrial solvent



Using a chemical solvent in an open container in a poorly ventilated area.

2 Sharp edge



A sharp edge on a metal sheet.

3 Electrical socket



An overloaded electrical socket.

4 Working at height



A person using a ladder to work at height.

5 Hot surface



A hot cooking surface in a kitchen.

6 Moving machinery



A rotating machine part without a guard.

7 Slippery floor



A wet and slippery floor in a walkway.

8 Sun exposure



Working outdoors under strong sunlight for a long time.

Your Risk Assessment Table (Filled Example)

No.	Situation	Hazard (what can cause harm?)	How can exposure happen?	Possible harm (what could happen?)	Likelihood (Low / Med / High) + reason	Severity (Low / Med / High) + reason	Overall risk (Low / Med / High)	Incident or Accident?	Control measure (optional)
1	Industrial solvent	Chemical solvent (vapors)	Inhalation of vapors	Respiratory irritation/ long-term health problems	High (open container, poor ventilation)	High (serious health effect)	High	Incident (if spill but no harm) Accident (if illness occurs)	Keep container closed, use ventilation, PPE (mask)
2	Sharp edge	Sharp metal edge	Touching the edge	Cut to hand	Medium (if handled)	Medium (moderate injury)	Medium	Accident (if cut occurs)	Use gloves, cover edge, debur the metal
3	Electrical socket	Electricity (overload)	Using multiple devices	Electric shock or fire	Medium (happens often)	High (serious injury or fire)	High	Incident (sparking) Accident (fire or injury)	Avoid overloading, use proper sockets, RCD
4	Working at height	Gravity – Working at high	Using a ladder	Fractures or serious injury	Medium (if no proper safety)	High (serious injury)	High	Accident (if fall occurs)	Use correct ladder, harness, follow safety procedures
5	Hot surface	Hot stove/pan	Touching hot surface	Burns	High (common in use)	Medium (moderate injury)	High	Accident (if burn occurs)	Use heat-resistant gloves, warning signs
6	Moving machinery	Rotating machine part	Clothing or hand gets caught	Crush injury/amputation	Low (if rarely accessed)	High (very serious injury)	Medium	Accident (if caught)	Use guard, keep safe distance, training
7	Slippery floor	Wet floor (low friction)	Walking on wet surface	Fall and sprain/fracture	Medium (happens sometimes)	Medium (moderate injury)	Medium	Accident (if fall occurs)	Clean spills, use warning sign, non-slip flooring
8	Sun exposure	UV radiation from sun	Working outdoors for long time	Sunburn, heat exhaustion, long-term skin damage	High (long time exposure)	Medium (moderate injury)	High	Incident (heat stress symptoms) Accident (heat stroke)	Use sunscreen, hat, shade, rest breaks, hydration

Answers to the questions (example answers)

- 1 The highest risk: Industrial solvent / Working at height / Sun exposure (all High). Because they have high likelihood and/or high severity.
- 2 The lowest risk: Moving machinery (Medium) or Sharp edge (Medium/Low depending on control measures).
- 3 Yes, the hazard can stay the same while this risk changes. Example: The same industrial solvent is lower risk when stored in a closed, ventilated container, but higher risk when opened in a poorly ventilated area.

- 4 The most effective control measure for the highest risk depends on the situation:
 - For industrial solvent: ventilation + closed container + PPE.
 - For working at height: proper ladder/harness.
 - For sun exposure: sunscreen, shade, rest breaks.
- 5 I learned that risk depends on both likelihood and severity, and the same hazard can lead to different levels of risk depending on the conditions and controls.



Submission

You can submit your work as:

- A completed table (Word / PDF / handwritten)
- OR a short presentation (3–5 slides)
- Include pictures and simple explanations.

Thanks



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Unit 1 (TC1 - Occupational Health & Safety)

Lesson 01 • Subtopic 1.3

Prevention : Act Before Harm Occurs

ENG. Nada Ahmed Ali

Date/4/10/2026

Academic Year:
Semester :

2026/2027
One

Recap

1- What is the hazard: the sharp metal edge, or the cut it may cause?

**Sharp metal
edge**

The cut

**Both are
the hazard**

Recap

-2A worker slips on a wet floor but stays on their feet. Incident or accident?

Incident

Accident

*Once we know the hazard — what should we do about it **BEFORE** it causes harm?*

Prevention — Act Before Harm Occurs

D

Prevention is the set of measures, procedures and organisational arrangements used to anticipate a harmful event and reduce the likelihood that it will occur.

KEY MESSAGE

Prevention means acting before the harmful event occurs.

Prevention therefore focuses on acting before harm happens, not reacting after it.

:Preventive Action Can Involve

- 1 **ELIMINATION** a source of danger where this is possible
- 2 **SUBSTITUTION** something hazardous with a safer alternative
- 3 **ENGINEERING CONTROLS** exposure to the source
- 4 **ADMINISTRATIVE CONTROLS** change the way people work
- 5 **PERSONAL PROTECTIVE EQUIPMENT** to protect the worker



Before You Begin a Task, Ask:

- 1 WHAT COULD CAUSE HARM?** Identify the hazard.
- 2 WHO COULD BE EXPOSED?** Think about yourself and anyone else who may enter or work in the area.
- 3 HOW COULD EXPOSURE OCCUR?** Consider the task, the location and the condition of the equipment or material.
- 4 WHAT COULD THE CONSEQUENCE BE?** Identify the possible injury or ill health.
- 5 WHAT SHOULD BE DONE BEFORE WORK CONTINUES?** Consider what preventive action is required.

This is the beginning of professional safety thinking.

RISK ASSESSMENT

- 1 IDENTIFY HAZARDS
- 2 WHO COULD BE HARMED?
- 3 ASSESS THE RISK
- 4 IMPEMENT CONTROLS
- 5 RECORD AND REVIEW



Putting the Three Core Terms Together

Term	What it tells you	Example
Hazard	What has the potential to cause harm	Sharp metal edge
Risk	How likely harm is under the conditions present, and how serious it could be	High possibility of hand contact while manually handling an exposed sheet edge
Prevention	What can be done to stop or reduce the chance of the event	Prevent access to the sharp edge, or remove/control the hazardous condition before handling

? Did You Know



A workplace does not become safe simply because an accident has not happened yet.

A hazard may remain unnoticed for a long period before the conditions arise that allow it to cause harm.

This is why hazard recognition and prevention must take place before an incident.

Industry Example — Sharp Metal Edge

SITUATION

A worker needs to move materials close to a sheet-metal storage area.



Before continuing, work through the five prevention questions for this scenario.

Working Through the Five Steps

- 1 Identify the hazard
- 2 Identify the exposure
- 3 Identify the harm
- 4 Think about the risk
- 5 Think preventively

PARTECIPATE
TOGETHER



You have moved from “That is dangerous” to explaining what the hazard is, how exposure could occur, and what harm might result.

Working Through the Five Steps

- 1 Identify the hazard** The sharp metal edge is the hazard.
- 2 Identify the exposure** A worker could contact the edge while walking past or handling nearby material.
- 3 Identify the harm** Possible harm includes a cut or laceration.
- 4 Think about the risk** Contact is more likely because the edge projects into an area where people are working.
- 5 Think preventively** The unsafe condition should be controlled before normal work continues.

You have moved from “That is dangerous” to explaining what the hazard is, how exposure could occur, and what harm might result.

Recognizing a Hazard ≠ Being Authorized to Fix It

You CAN

- Observe
- Identify
- Warn
- Record
- Report

You must NOT (without authorization)

- Repair
- Adjust
- Remove
- Isolate
- Operate
- Certify

REJECTED: “I would just bend the metal back”.

Same Hazard, Different Risk – Page 18 —HOME WORK

A container of industrial solvent, under two conditions:

A
Closed container, stored correctly, in a ventilated area, protected from unnecessary access.

B
Same product open, poor ventilation, no suitable controls.

- .1Is the chemical hazard different between A and B?
- .2What condition has changed?
- .3In which situation is exposure more likely?
- .4In which situation is the risk greater?
- .5Suggest one preventive action.

— Say It Professionally 1.1 Activity

Rewrite this weak statement so it clearly identifies the hazard, exposure and possible harm:

“The sharp metal is dangerous”.

Your improved statement: